

CROSS-MODAL REPRESENTATIONAL ALIGNMENT IN VISION AND AUDIO MODELS

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Abstract

The Platonic Representation Hypothesis proposes that independently trained neural networks converge toward shared internal representations [1]. While vision–language and audio–language alignment are well-established [1, 2, 3], direct vision–audio convergence without cross-modal training or linguistic intermediary remains underexplored, limiting understanding of whether convergence reflects shared world structure or language-mediated semantics. Using 620 semantically paired clips and images from AudioCaps2.0, layer-wise embeddings are extracted from DINOv2-large/base (vision), BEATs-iter3/iter3+ (audio), and Qwen3-1.7B (language), measuring alignment with RSA and linear predictivity across model combinations and layer pairs. Results reveal significant vision–audio alignment in mid-to-late layers (peak LP $r \approx 0.295$), though consistently weaker than vision–language ($r = 0.402$) and audio–language ($r = 0.357$) alignment. These findings support cross-modal convergence between independently trained sensory models while raising broader questions about whether representational convergence reflects language-like semantic abstraction, direct world structure, or a mixture of both.

Introduction

A central question in cognitive neuroscience and machine learning is how distinct sensory systems develop shared internal representations of the world. Biological vision and audition converge into unified perception despite relying on different neural pathways in the human brain, and the same question now arises in deep learning.

The Platonic Representation Hypothesis [1] proposes that models converge toward shared representations regardless of modality, and prior work has established alignment between vision-language [1, 2] and audio–language models [3]. However, direct vision–audio alignment without any cross-modal training or language intermediary remains largely unexamined.

This study directly tests this using 620 semantically paired audio clips and images from AudioCaps 2.0, comparing layer-wise representations from DINOv2 (vision) [4], BEATs (audio) [5], and Qwen3-1.7B (language) [6].

This study addresses the following research questions:

1. Do independently trained vision and audio models show representational alignment on semantically paired inputs, and does this alignment follow the early-to-late progression observed in vision-language systems?
2. Is vision-audio alignment stronger or weaker than vision-language and audio-language alignment?
3. Does vision model capacity affect cross-modal alignment, such that larger models show stronger convergence with audio representations?
4. Do audio models trained with labeled data show stronger cross-modal alignment than those trained without labels?
5. Do images generated from expanded audio captions show greater alignment with audio than naturalistic images?

Methods

Models:

Model	Modality	Params	Layers	Hidden Dim	Tokenizer	Output Shape
DINOv2-large	Vision	307M	24	1024	Self-supervised	(25, 1024)
DINOv2-base	Vision	86M	12	768	Self-supervised	(13, 768)
BEATs-iter3+	Audio	90M	12	768	Supervised	(12, 768)
BEATs-iter3	Audio	90M	12	768	Self-supervised	(12, 768)
Qwen3-1.7B	Language	1.7B	29	2048	—	(29, 2048)

Tab. 1: Information about selected models.

Dataset:

AudioCaps2.0 [7] contains audio clips sourced from YouTube paired with natural language captions. The following components were used for the analyses:

- **Audio:** raw files received from AudioCaps researchers; filtered out audio clips less than 8 seconds (mean duration = 9.9 seconds)
- **Middle Frames:** center image from random subset of raw AudioCaps video files; manually filtered to remove blank images, animated video, etc.
- **Generated Images:** expanded captions from selected middle frames to include more visual cues (using Microsoft-4k-phi model [8]) for generating images using Stable Diffusion XL [9]
- **Captions:** 620 captions corresponding to audio files of videos used to extract middle frames

Alignment Metrics

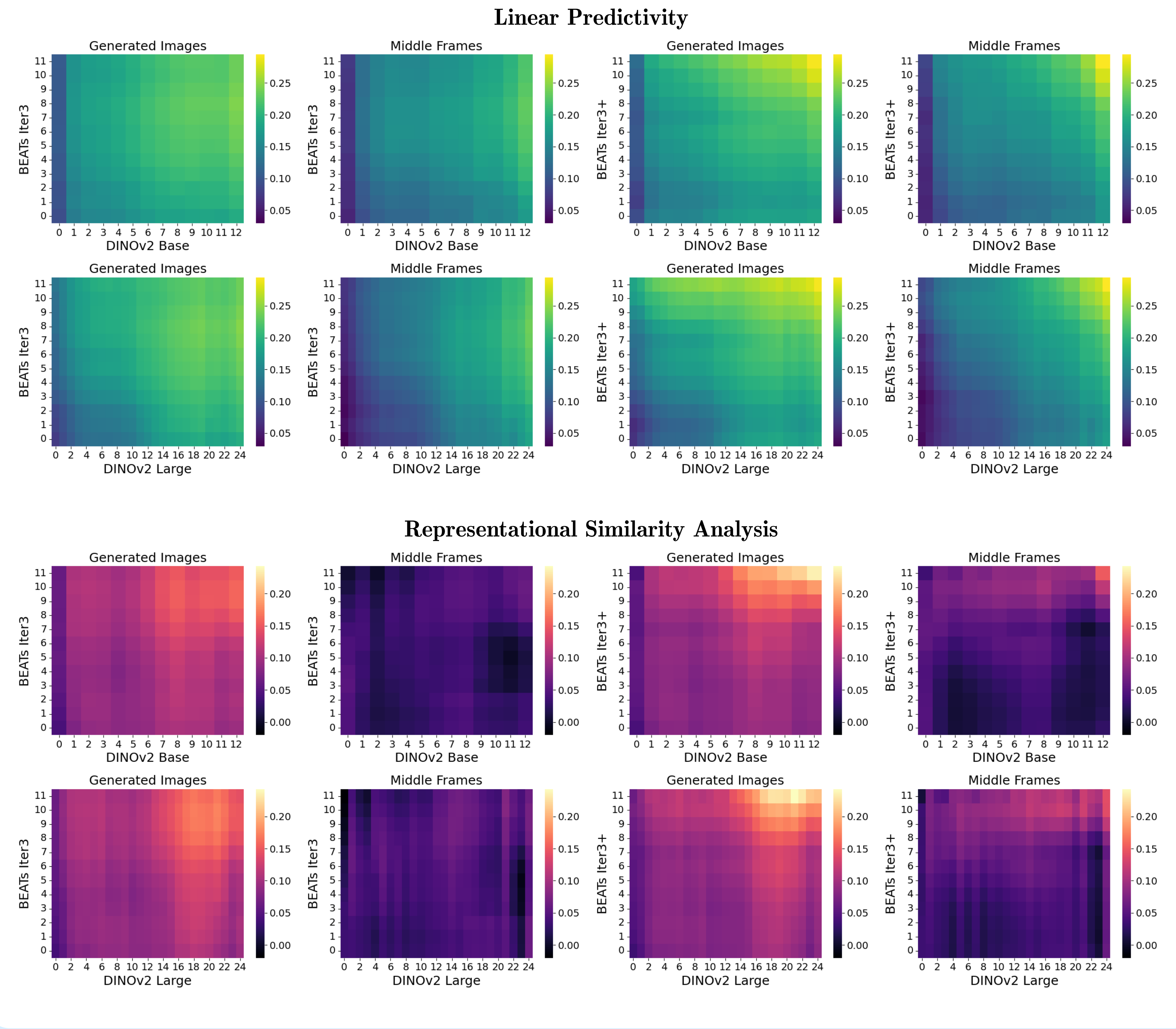
Linear Predictivity:

Linear predictivity [10] measures how well representations from one modality linearly predict those of another. Ridge regression with cross-validated regularization is used to map embeddings across all layer pairs, estimated in both directions and averaged. Alignment is reported as the average pairwise correlation between predicted and true feature values across test sets in a 5-fold cross-validation scheme.

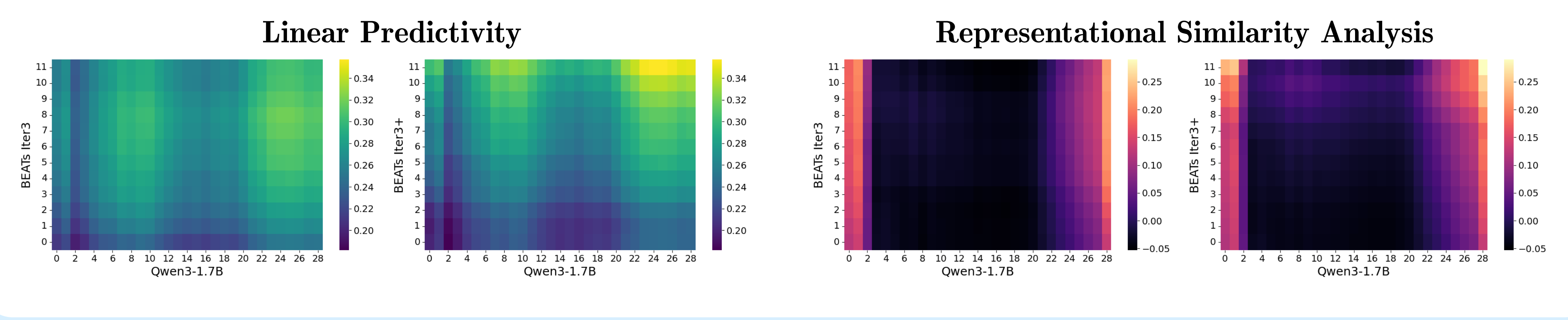
Representational Similarity Analysis (RSA):

RSA [11] compares representational geometry across modalities without requiring a shared embedding space. For each combination of model layers, a Representational Dissimilarity Matrix (RDM) is constructed using correlation as the distance metric, with embeddings z-score normalized per layer. RDMs are compared using Spearman rank correlation between their upper triangles.

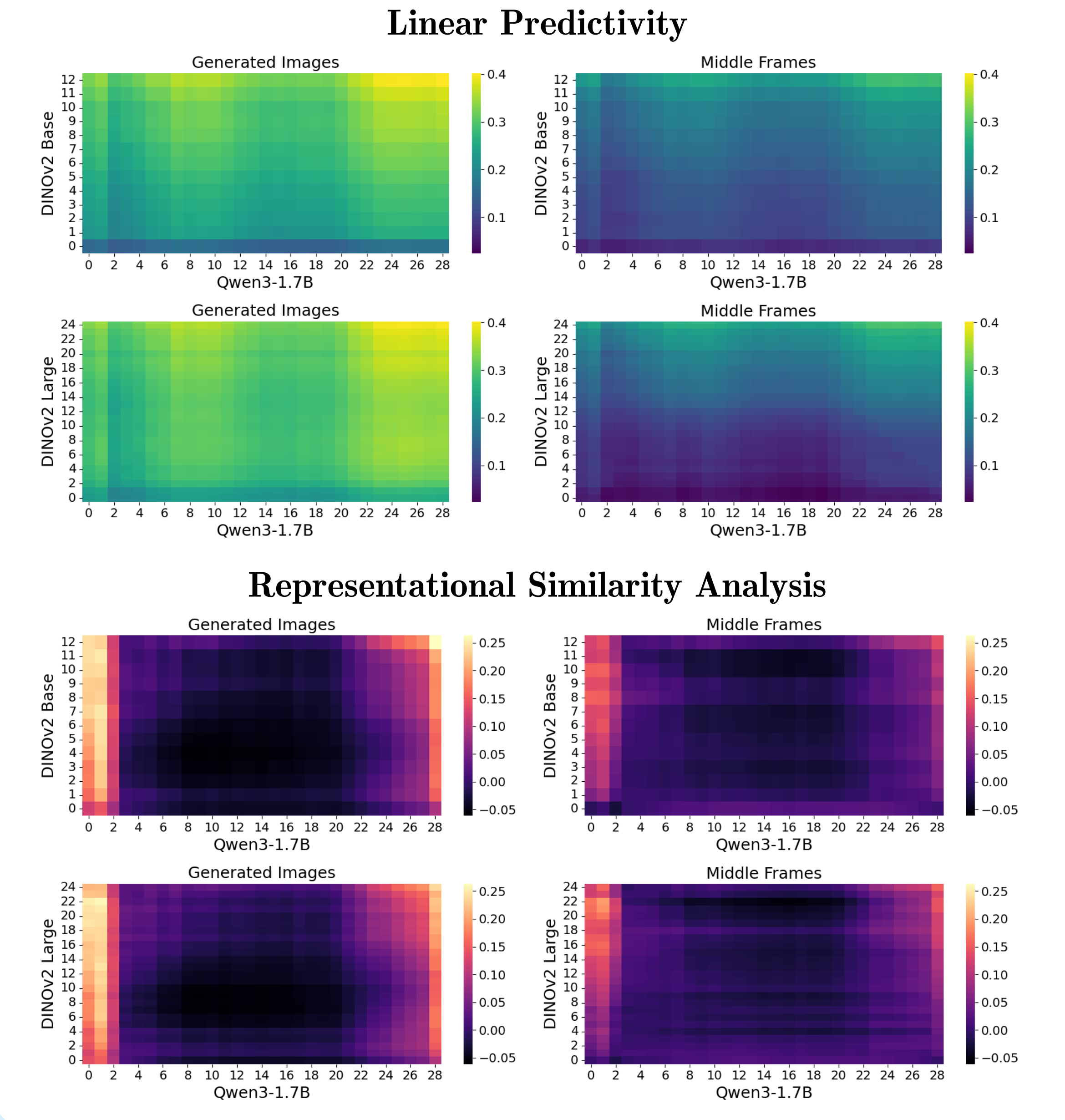
Results: Vision-Audio Alignment



Results: Language-Audio Alignment



Results: Language-Vision Alignment



Conclusion

- This study provides evidence that independently trained vision and audio models develop **meaningful representational alignment** on semantically paired inputs, with convergence emerging primarily in **mid-to-late network layers**, consistent with patterns observed in vision-language systems though more broadly distributed across layers.
- **Vision-audio alignment remained the weakest of the three modality pairs** (peak LP $r = 0.295$, RSA $\rho = 0.243$), positioning language as the stronger representational bridge, though this may partly reflect language encoding semantic regularities associated with both modalities rather than direct cross-modal mediation.
- **Supervised audio pre-training** (BEATs-iter3+) consistently improved alignment across both vision model sizes ($\Delta \approx +0.055$, whereas **vision model capacity** showed only modest, inconsistent effects, providing limited support for the Platonic Representational Hypothesis prediction that larger models converge more strongly).
- **Caption-conditioned generated images** produced stronger early-layer alignment than naturalistic middle frames, though both converged in later layers, reflecting that generated images may be more **semantically regularized** rather than indicating a general advantage of synthetic stimuli over temporally samples natural frames.
- Together, these findings suggest that cross-modal convergence between vision and audio reflects a mixture of **shared world structure** and **language-mediated semantics**, with practical implications for stimulus construction and model selection in future multimodal alignment research.

References

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